Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/11/2005
PI Name:	Pollack, Lois Ph.D.		
Project Title:	Microscale Mixer for Protein Folding		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Biological physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	14853	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	08/01/2001	**End Date:**	03/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Batt, Carl A. (Cornell University) Batt, Carl A. (Cornell University)		
Grant/Contract No.:	NAG3-2942		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the un iverse obscured by Earth's gravity		

The broad, long-term objective of this work is to understand the interactions involved in macromolecular folding and function. It is necessary to trigger and monitor structural changes on time scales as short as microseconds to observe these interesting dynamics. At present, our understanding is limited by the lack of suitable experimental tools accessing dynamics on time scales ranging from microseconds to seconds. In our experiments, we employ a microscale fast fluid mixer with two different probes of macromolecular structure: small angle x-ray scattering to determine global size and shape changes as well as high sensitivity fluoresence microscopy to probe more local interactions and dynamics. The flow cells are compact, and may be adaptable to a gravity-free environment. In previous versions of these flow cells, the macromolecules were in direct contact with the walls defining the micro-flow channels. As a result, some aggregation of protein occured on the surfaces, causing a small, but persistent, background increase as a function of

Task Description:	time. In our proposed flow cell design, a layer of buffer solution flows along all of the surface and prevents their contact with the folding macromolecules. We are also exploring the use of different materials for the cells and windows. Once complete, these new flow cells will permit higher quality measurements by decreasing (or eliminating) extraneous sources of scatter. With this improved signal to background ratio, better quality data can be acquired from a smaller quantity of sample, a critical consideration when dealing with the planned study of mutants. The objective of this project is to study global structural changes that occur during the initial, rapid steps of macromolecular folding. The technique that we have developed monitors rapidly varying molecular conformations by synchrotron x-ray scattering. Sharp time resolution is achieved by the use of a microfabricated flow cell. This technology will be applied to study the folding of both wild-type and variant macromolecules. The latter will be carefully selected to elucidate the role of specific residues in the formation of native, compact structures. Fluorescence microscopy will also be used to monitor local dynamics. A long term goal is the application of both techniques to the same molecule, to acquire both local and global structural information. Our understanding of the dynamics of conformational change in macromolecules is limited by a lack of suitable, real-time experimental probes of their most basic parameters: size and shape. The goal of this project is to probe the dramatic changes that occur at the very initiation of the folding process, specifically the collapse of the flexible, denatured chain into a compact structure. In addition, we have begun experiments to measure the solution environment around DNA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	An unfortunate by-product of space travel is the ubiquitous, high level of background radiation. It is well known that DNA is susceptible to radiation damage. At least one mechanism for damage involves interactions with the solution that immediately surrounds the DNA, once the entire system has been exposed. It is therefore important to have an accurate representation of DNA's solution environment. This past year, in addition to verifying that the spatial distribution of ions is well-represented by an electrostatic model (R. Das, T. T. Mills, L. W. Kwok, G. S. Maskel, I.S. Millett, D. Doniach, K. D. Finkelstein, D. Herschlag and L. Pollack, "Counterion distribution around DNA probed by x-ray scattering" Phys. Rev. Lett., 90, 188103 (2003)), we have successfully measured the concentration of ions at the surface of the DNA, setting the scale factor for the ion distribution (K. Andresen, R. Das, H.Y. Park, L.W. Kwok, J.S. Lamb, H. Smith, D. Herschlag, K.D. Finkelstein and L. Pollack "Spatial distribution of competing ions around DNA", Phys. Rev. Lett., 93, 248103, (2004).).
Bibliography Type:	Description: (Last Updated: 07/19/2007)
Articles in Peer-reviewed Journals	Andresen, K., Das., R., Park, H.Y., Smith, H., Kwok, L.W., Lamb, J.S., Kirkland, E., Herschlag, D., Finkelstein, K., and Pollack, L. "Spatial Distribution of Competing Ions around DNA in Solution" Physical Review Letters , Dec-2004